

Name of the student Deesham JainScholar no 24U030008**Indian Institute of Information Technology Bhopal****Department of Information Technology****Mid Term Examination – September 2025****Subject Code:-** IT-3002**Semester-** IIIrd**Subject Name:-** Database Management System**Time:** 10:00AM- 11:30AM**Max Marks:** 20**Note:** All questions are compulsory and carrying equal marks.

S. No.	Question	MM
1	Consider the following relation schema $R(A, B, C, D, E)$ with the set of functional dependencies: $A \rightarrow B$ $B \rightarrow C$ $A \rightarrow D$ $D \rightarrow E$ Normalize the given relation schema R to 1NF, 2NF, and 3NF, explaining the rationale for each step. After achieving 3NF, further normalize the relation to Boyce-Codd Normal Form (BCNF). Identify and resolve any violations of BCNF in the relation, providing justification for each step.	5
2	(a) Consider the relation schema $R(X, Y, Z, W, V)$ with the following set of functional dependencies: $\{X \rightarrow YZ, XY \rightarrow WV, Z \rightarrow W, W \rightarrow XV\}$. Identify the candidate keys for R and compute the canonical cover of the given set of functional dependencies. (b) Consider a relation $R(A, B, C, D, E, F)$ with following set of functional dependencies holding on it: $\{AB \rightarrow C, AD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. Find the highest normal form of relation R . Synthesize losslessjoin dependency preserving 3NF relations from R .	5
3	(a) Codd's Rule 3 (Systematic Treatment of Null Values) ensures that nulls are treated in a uniform manner as "missing" or "inapplicable" data. Suppose in a hospital database, NULL is used inconsistently — sometimes for "unknown blood group" and sometimes for "not tested." - What anomalies can arise in query processing and reporting? - Suggest how strict adherence to Rule 3 could resolve these issues. (b) Codd's Rule 1 (Information Rule) states that all information in a relational database should be represented explicitly at the logical level using tables. Imagine a library database where some metadata (e.g., number of borrowed books) is maintained only in application code rather than in tables. - Analyze how this violates Rule 1. - Discuss potential risks to consistency and portability of the system.	5
4	A hospital database requires the following tables: - Patient: Patient ID, Name, Date of Birth, Gender (M/F), Blood Group (must be from $\{A^+, A^-, B^+, B^-, AB^+, AB^-, O^+, O^-\}$). - Doctor: Doctor ID, Name, Specialization, Salary ($\geq 50,000$). - Appointment: Appointment ID, Patient ID, Doctor ID, Date, Time. - Prescription: Prescription ID, Appointment ID, Medicine, Dosage, Number of Days (1–30). Write SQL queries to: a) Create these tables with referential integrity (foreign keys). b) Add check constraints for salary, gender, blood group, and dosage. c) Ensure each appointment links a valid patient and doctor.	5